

Frédéric Kah

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Summary

Engineering student at EURECOM (Embedded Systems), currently on exchange at DTU, working on GPU programming, performance optimisation and hardware design. Co-developed a GPU GEMM kernel that reaches 87% of cuBLAS, and had a contribution merged into the open-source fuzzer LibAFL. Seeking a **6-month end-of-studies internship in Paris starting early February 2027** in GPU/CPU performance or systems engineering.

Projects

High-Performance GEMM in CUDA & HIP

03/2026 – 07/2026

Two-person project, co-author · github.com/Surypp/gemm · Technologies: CUDA, PTX, HIP, C++, Nsight Compute

- Co-developed an FP16/FP8 GEMM from scratch in CUDA and inline PTX, optimising it step by step with Nsight Compute: shared-memory tiling, tensor cores, asynchronous memory pipelining (`cp.async`, TMA) and FP8 on Blackwell. Reached **104.5 TFLOPS (86.8% of cuBLAS)** on 8192×8192×8192 FP16 on an RTX 5080.
- Led the **AMD HIP port** (RX 9060 XT) up to double-buffered matrix-core kernels, reaching **about 50% of rocBLAS**; ported the CUDA benchmark harness to NVIDIA A100, GH200 and RTX 5080.

GPU-accelerated DNA Data Encoding for OligoArchive

09/2025 – 02/2026

Supervisor: Raja Appuswamy (EURECOM) · Technologies: CUDA, C++, TBB

- Ported the lookup-table generation of a TBB-parallelised CPU encoder/decoder to CUDA, optimising memory layout and batching, with a compatibility layer keeping the CPU version working. **40× speedup** on two RTX 2080 Ti GPUs: the table for 22-nucleotide sequences (longer ones give higher compression) now takes a few hours instead of a week.

Systems & Fuzzing Research

09/2025 – 07/2026

Supervisors: Aurélien Francillon, Marco Cavenati (EURECOM) · Technologies: Rust, C, Python, LibAFL, QEMU

- Intel PT on Windows (Rust)** 09/2025 – 01/2026
Added Windows support for Intel PT hardware tracing to LibAFL, the open-source fuzzing library from the AFL++ team (2.6k GitHub stars), which previously had none. Built a Rust interface to the undocumented Windows Intel PT driver, enabling coverage-guided fuzzing of closed-source Windows binaries; merged upstream (PR #3845).
- QEMU snapshots (Python)** 02/2026 – 07/2026
Wrote a Python benchmarking harness that drives QEMU VMs through the GDB stub and guest agent, varies guest RAM and working-set size, and collects per-restore CPU, I/O and page-fault metrics from Linux `/proc`, with 95% confidence intervals. Used it to compare five snapshot-restore strategies for fuzzing and determine which one fits best depending on how many pages each run modifies.

Parametrisable TPU in Chisel

09/2026 – present (planned end: 12/2026)

Supervisor: Martin Schoeberl (DTU) · Technologies: Chisel, Scala, FPGA

- Designing a TPU-style matrix accelerator in Chisel, parametrisable in array size, data width and dataflow, targeting validation on FPGA.

Work Experience

Software Developer Intern, Worldline

06/2025 – 07/2025

Technologies: Python, SQL

Built a tool that rewrites Oracle SQL queries into PostgreSQL-compatible ones using static analysis and pattern-based transformations; it translates **94% of the queries** automatically.

Education

Exchange Semester, DTU (Technical University of Denmark), Copenhagen

09/2026 – present (ends 01/2027)

Chosen to build skills in high-performance computing, parallel architectures and hardware design.

Courses: Agile Hardware Design • High-Performance Computing • Parallel Computers • Large-Scale Modelling

Engineering Degree (Master level), EURECOM, Sophia Antipolis, France

09/2024 – present

Specialisation: Embedded Systems · Expected graduation: 09/2027

Relevant courses: Computer Architecture • Operating Systems • Digital Systems & HW/SW Integration • Database Management Systems • Distributed Systems

Preparatory Classes (MPSI and MP), Lycée Charlemagne, Paris, France

09/2022 – 06/2024

Technical Skills

Programming Languages	C, C++, Rust, Python, Scala, RISC-V Assembly, Haskell, OCaml
GPU & Performance	CUDA, HIP/ROCm, PTX, MPI, Nsight Compute, perf
Hardware	Chisel (Scala HDL), VHDL, FPGA, hardware accelerator design
Systems	Linux, Windows internals, QEMU, Intel PT, LibAFL
Tools	Git, CMake, Docker, LaTeX

Languages

French (native) · English (C1, TOEIC 975/990) · Italian (A2)